Exp:9 Date:

DESIGN A SIMPLE TOPOLOGY AND CONFIGURE WITH ONE ROUTER, TWO SWITCHES AND PCs Using CISCO PACKET TRACKER

Aim: To design and configure a simple network topology consisting of one router, two switches & multiple PC's in cisco packet tracer, and enable communication between PCs.

Definitions:

* Routers- A networking device that forwards data packets blw computer networks, directing traffic on internet

* switch = A device that connects devices within a network and uses a MAC addresses to forward data to the correct destination.

* Pc (personal computer): End devices in a network that users interact with to send and received data.

* cables:
(i) straight through cable: Used to connect diff devices like PC to switch or switch to Router.

[ii] Crossover cable: Used to connect similar devices like switch to switch [rarely needed with modern switches]

[iii] Console cable: Used to connect PC to router or switch console port for configuration.

Equipment Needed:-

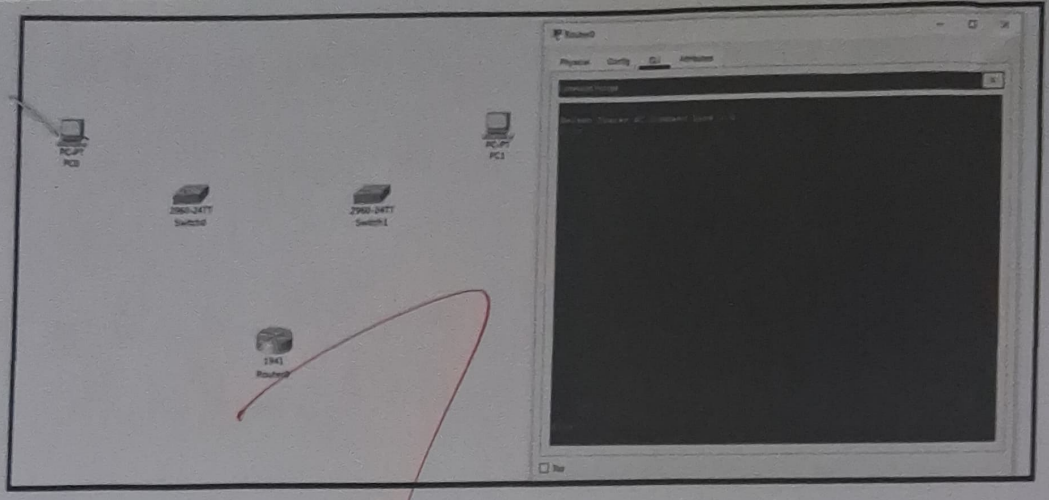
* Cisco packet Tracer software
* 1 Router (eg: 2811)
* 2 Switches (eg: 2960)
* 4 PC's
* Appropriate cables (copper straight-thro

Step-By-Step procedure:-

step 1: open cisco packet tracker
* Launch cisco packet tracker on your comp

Step 2: Add devices to workshop:-
* From the bottom device toolbar, drag & drop

drop :- * 1 Router (1941)
* 2 Switches (2960)
* 4 PC's (2 connected to each sw



fast ethernet 0/1.

PCs to the switches

*PC0 to switch 0 Fast Ethernet 0/2
*PC1 to switch 0 Fast Ethernet 0/3
* PC_2 to switch 1 Fast Ethernet 0/2
* PC_3 to switch 1 Fast Ethernet 0/3

Step 4: configure Router interface
*click the router
*click tab
* Enter the IP address & Give no shutdown command

Step 5: Configure Pc IP addresses
*click on Each Pc
*Desktop tab
*IP configuration & Assign IP addresses & Gateways

For Eg: PC0. 192.168.1.10
255.255.255.0

step 6: Save and Test connectivity

* Save the packet Tracer File
* Test the connectivity using the Ping command From PC0 to PC_2 or PC_3
* Go to CMD type ping: 192.168.2.10
* you should see replies if the network is connected successfully

Result: The PCs connected to different switches can communicate with each other via the router. The successful Ping replies verify that the network is properly configured & operational.